Second version of the Powered Platter design

V1 was functional and came to gether well. It had a few key flaws that had to be addressed

- The microcontroller used did not have adequate memory to control the neopixels as I wanted I quickly ran into memory issues and limitations that are not worth the tiny form factor

- The small traces did resolve when milling with the CNC, however, traces that were thin and long were very easily damaged when soldering. This caused extra bodge wires and was an unpleasant assembly experience. Increase size of traces whenever possible

- This version is not going to include Neopizels, even though the upgraded microcontroller included in this version has the memory to support them. I'm replacing the Neoplades with LED blades that use 3mm through-hole LEDs. This increases the connections and limits the color output of the generated display. The benefit is simplified timing.
  - Consider breaking out a connection for the LEDs if possible

V2 includes an upgraded microcontroller - STM32C071KBT6 - with much more memory and I/O.

Hardware notes

- V1 was too thick for the standard thumb screw
- cuttout for wires below was not deep enough

MCU

Power

Connectors

level shifting

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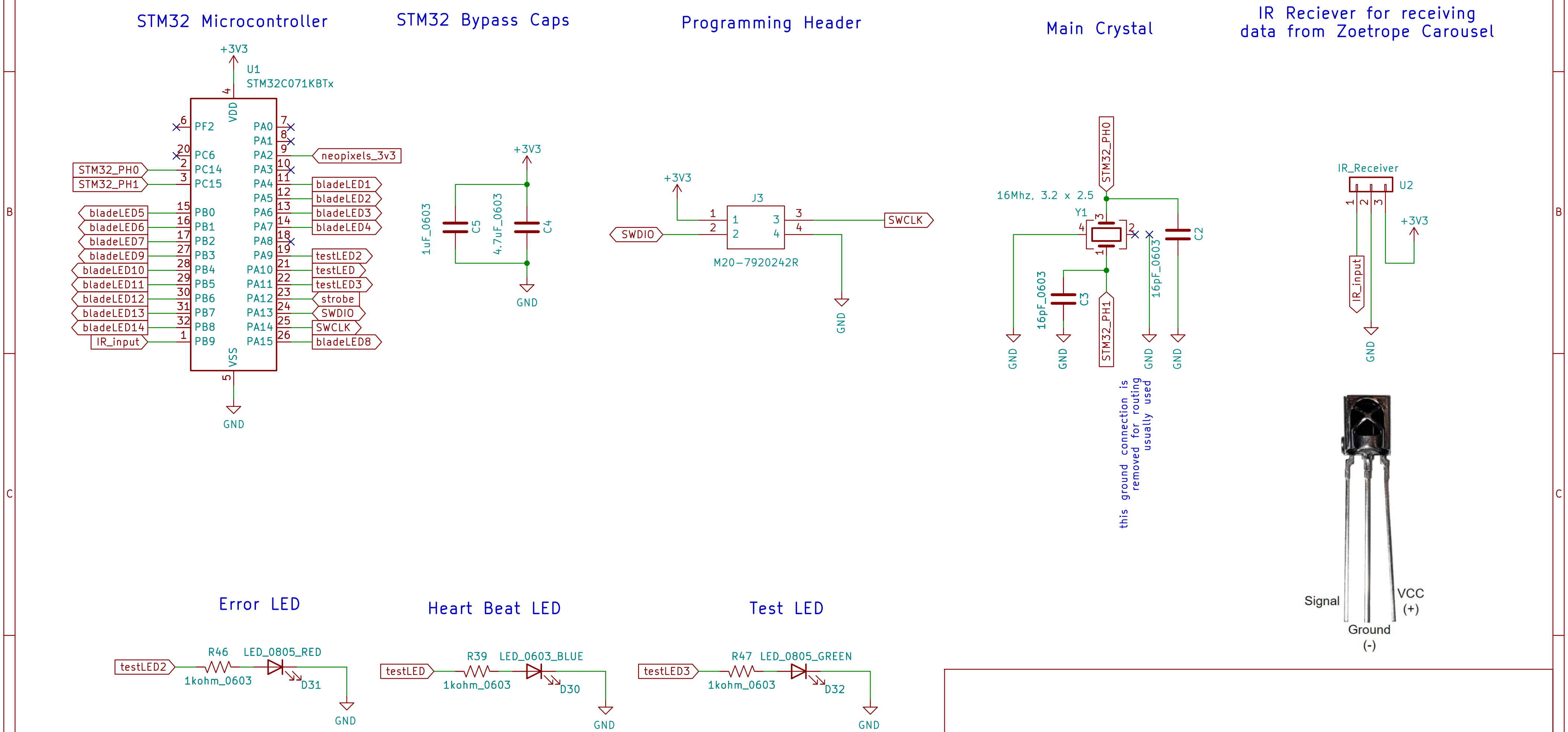


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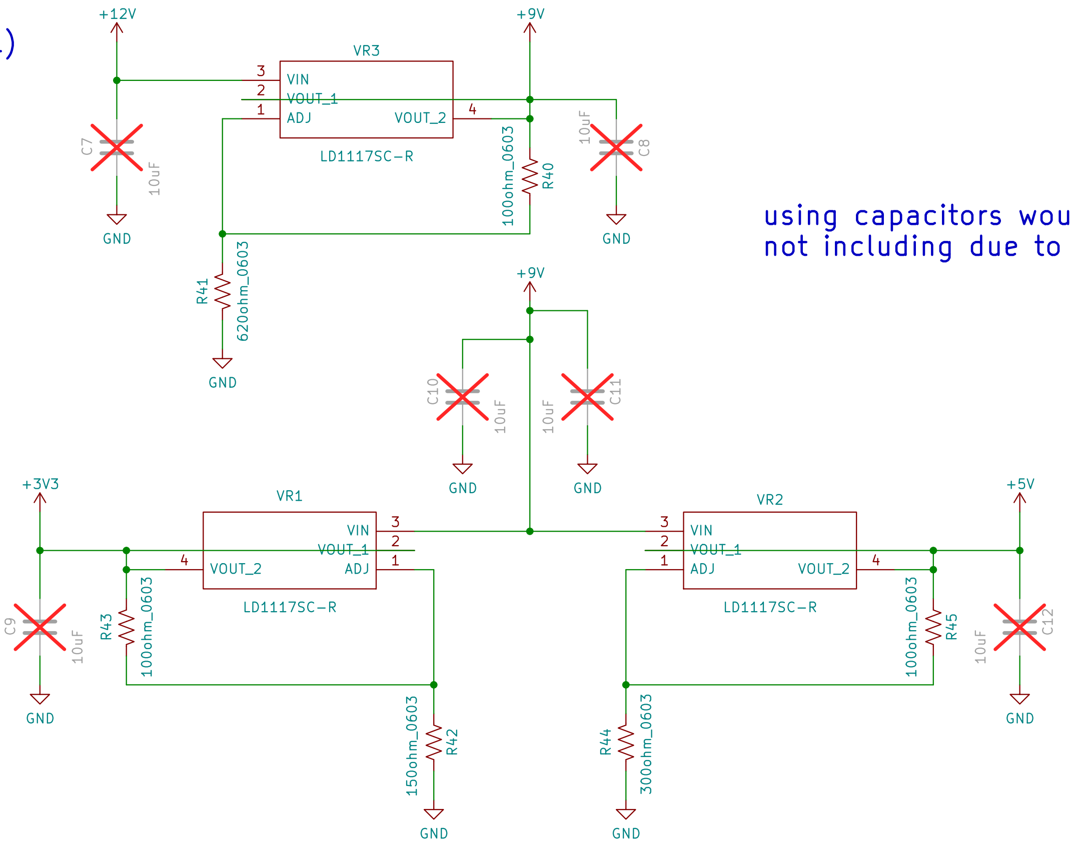
Three LDOs for power to the Platter

9VDC LDO takes the 12V to 9V, this is here for heat reasons – bypass available in case its not necessary
5VDC LDO for the Neopixels and index/speed sensor
3VDC for MCU

Vout = Vref(1 + R2/R1)
Vref = 1.25
R1 = 100




using capacitors would improve ripple,
not including due to parts on hand

using 150 ohms sets the 3.3v
to 3.1v, which should be OK

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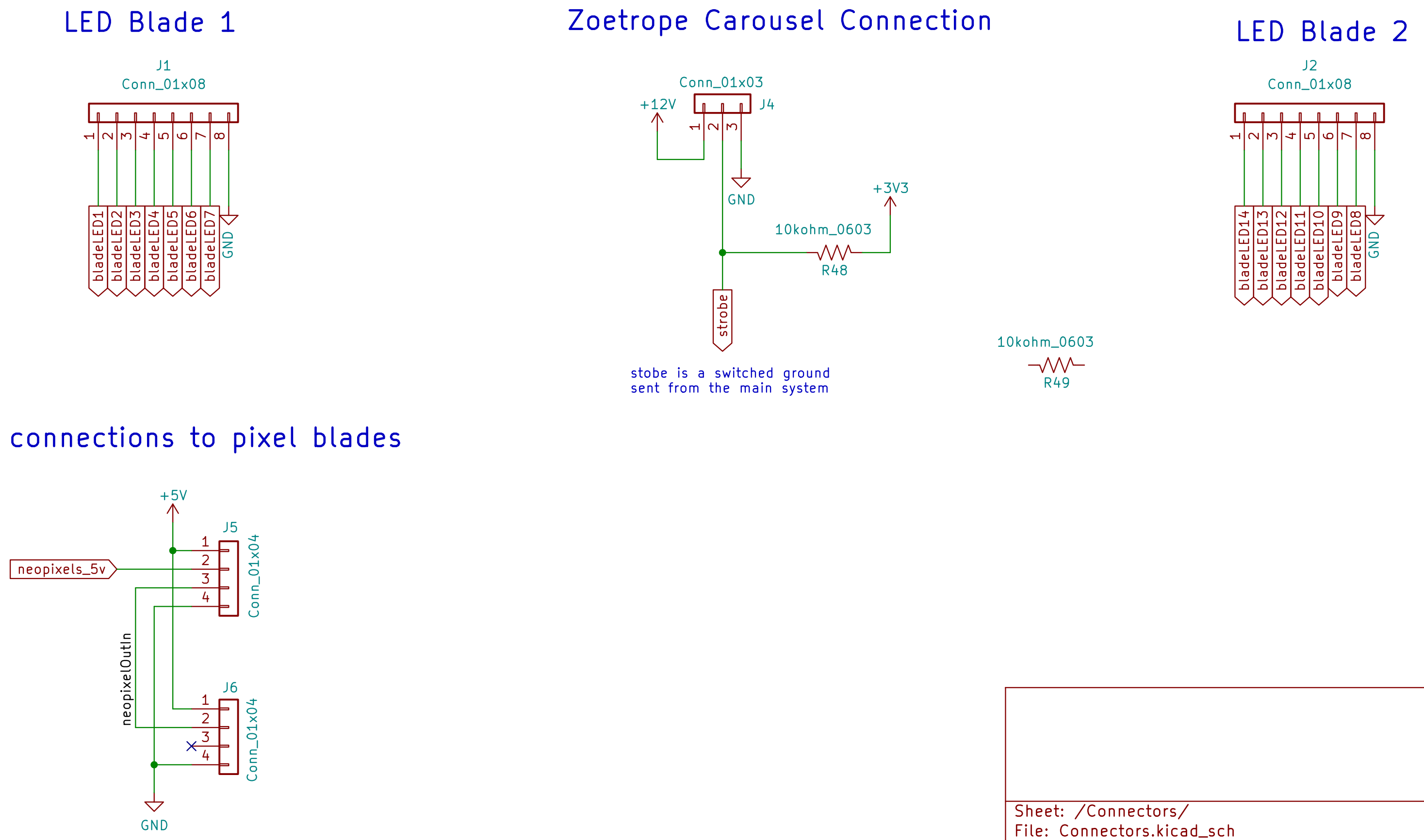


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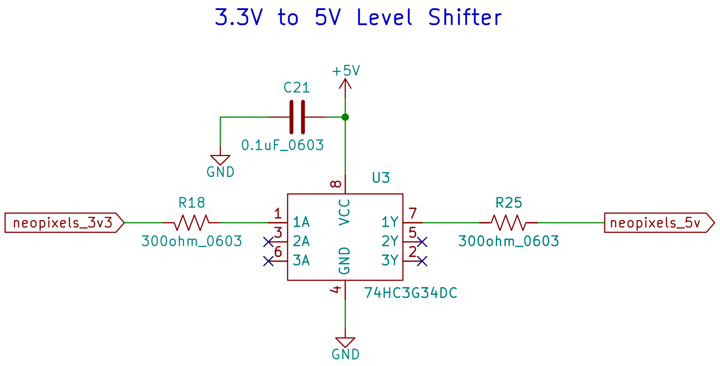


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